

TM4C123 LED Blinky — Stopwatch Extension

Implementation Report — EduARM4 Board (TM4C123GH6PM)

Objective

Extend the existing rate/colour LED blinky assignment with a stopwatch feature per the six assignment requirements: SysTick-based timekeeping, dual input (Serial + 4×4 keypad), a shared 7-segment display that switches between the stopwatch and the original RGB status readout, and interrupt-driven control for the onboard SW1/SW2 switches.

Implementation Summary

- Timekeeping: a SysTick interrupt ticks every 10 ms and drives both the stopwatch's elapsed-time counter (swCentis) and switch/keypad debounce timing — no busy-wait timing loops remain in the control path.
- Stopwatch state machine: DISABLED → IDLE → RUNNING ⇌ PAUSED, with enable/disable, start/stop, and pause/resume as three independent actions, each reachable from Serial and keypad alike.
- Serial console control: SWENABLE, SWSTART, SWPAUSE, and SWSTATUS commands, alongside the original RATE / COLOUR / PAUSE / RESUME / STATUS commands.
- Keypad control: row 0 (PE0) is held permanently low; columns PC4/PC5/PC6 map to keys '1', '2', '3' for enable/disable, start/stop, and pause/resume respectively, each firing a genuine falling-edge GPIO interrupt rather than a polled scan.
- Display sharing (enable/disable): when the stopwatch is enabled the 7-segment module shows elapsed time as MM:SS; when disabled it reverts to the original rate/colour/run-state readout, satisfying items 4 and 5 with a single updateDisplay() dispatcher.
- Interrupt-driven SW1/SW2: onboard switches are now handled through the GPIO Port F interrupt path rather than polling, preserving the original hold-to-increment and hold-both-to-pause behaviour.
- Vector table: SysTick_Handler and GPIOPortC_Handler are wired directly into the project's own startup file's vector table (entries 15 and 18), replacing an earlier RAM vector-table relocation workaround that is no longer needed.

Files Delivered

- main.c — application logic: LED blink/colour control, UART command parser, stopwatch state machine, keypad and switch interrupt handlers, 7-segment display dispatch.
- startup_ccs_patched.c — the project's CCS/GNU startup file with SysTick and GPIO Port C vector table entries pointed at the application's handlers.

Verification Notes

Recommended checks: (1) confirm RATE/COLOUR/PAUSE/RESUME/STATUS still behave as in the base assignment; (2) SWENABLE then SWSTART and confirm the display switches to a running MM:SS count; (3) SWPAUSE freezes the count, a second SWPAUSE resumes it; (4) SWENABLE again returns the display to the rate/colour/state readout while stopwatch time is retained in the background; (5) keys '1'/'2'/'3' on the keypad reproduce the same three actions independently of the serial console.